Numerical Physics I — Exercise 2

Pendulum with circular excitation, damping, resonance and chaos

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1 Introduction

In this work, we study the dynamics of a simple pendulum whose point of suspension undergoes a uniform circular motion. In addition to gravity, the model includes viscous damping and the inertial effects associated with the non-inertial reference frame attached to the moving support. As a result, the system exhibits a rich variety of behaviours depending on the forcing amplitude, the forcing frequency and the damping coefficient, ranging from small-amplitude oscillations to nonlinear resonance phenomena and chaotic motion.
The objectives of this exercise are twofold. First, we analyse analytically several limiting cases of the problem, in particular the small-oscillation regime without forcing and damping, as well as the dependence of the oscillation period on the amplitude for the nonlinear pendulum. Second, we solve the full time-dependent problem numerically using a second-order Verlet scheme adapted to accelerations depending not only on position, but also on velocity and time.
The report is organised as follows. Section 2 presents the analytical calculations. Section 3 describes the numerical model and its implementation in C++. Section 4 is devoted to the simulations and discussion of the results, including convergence tests, comparison with analytical predictions, resonance effects, sensitivity to initial conditions, and Poincaré sections associated with regular and chaotic regimes.

2 Analytical calculations

2.1 Equation of motion (a)

To derive the equation of motion we will work in the reference frame $\mathcal{R}'$ attached to the point O', which is in translation (but not rotation) with respect to the inertial frame. Therefore, Newton's second law can be applied by including the inertial force $-m\vec{a}_{O'}$.

To begin with, we define the position of the mass with respect to O', that is

$$\vec{r}_{m/O'} = L\sin\theta\,\vec{e}_x - L\cos\theta\,\vec{e}_y \tag{1}$$

The angular position of the pendulum is $\theta(t)$, measured from the vertical downward. Then we find the relative velocity in R' deriving the position over time.

$$\vec{v}' = \frac{d}{dt}\vec{r}_{m/O'} = L\dot{\theta}\cos\theta\,\vec{e}_x + L\dot{\theta}\sin\theta\,\vec{e}_y.$$

Its magnitude is $L\dot{\theta}$, and since the tangential direction is

$$\vec{e}_\theta = (\cos\theta, \sin\theta).$$

We end up with the following expression for the velocity

$$\vec{v}' = L\dot{\theta}\,\vec{e}_\theta. \tag{2}$$

Moreover, the relative acceleration is

$$\vec{a}' = L\ddot{\theta}\,\vec{e}_\theta - L\dot{\theta}^2\,\vec{e}_r,$$

and so the tangential component of the acceleration of the mass is simply

$$a'_\theta = L\ddot{\theta}. \tag{3}$$

To find the equation of motion we will be projecting the forces onto the tangential direction $\vec{e}_\theta$. These forces acting on the system are gravity, the viscous friction force ($\vec{F}_{\text{visc}} = -\kappa\vec{v}$) and the inertial force due to the motion of O'.

To begin with, the force due to gravity will be:

$$P_\theta = \vec{P} \cdot \vec{e}_\theta = -mg\sin\theta \tag{4}$$

For the friction force ($\vec{F}_{\rm visc} = -\kappa\vec{v}$), we first need to find the total velocity of the mass of the pendulum. That is taking into account the inertial term:

$$\vec{F}_{\rm visc} = -\kappa\vec{v} = -\kappa(\vec{v}_{O'} + \vec{v}') \tag{5}$$

The position of O' is

$$\vec{r}_{O'} = r\sin(\Omega t)\,\vec{e}_x - r\cos(\Omega t)\,\vec{e}_y$$

and its velocity is

$$\vec{v}_{O'} = r\Omega\cos(\Omega t)\,\vec{e}_x + r\Omega\sin(\Omega t)\,\vec{e}_y$$

The tangential component is

$$(\vec{v}_{O'})_\theta = \vec{v}_{O'} \cdot \vec{e}_\theta = r\Omega\cos(\Omega t)\cos\theta + r\Omega\sin(\Omega t)\sin\theta = r\Omega\cos(\Omega t - \theta) \tag{6}$$

and similarly,

$$(\vec{a}_{O'})_\theta = r\Omega^2\sin(\Omega t - \theta) \tag{7}$$

Thus, taking (2) and (6)

$$\vec{v} = \vec{v}_{O'} + \vec{v}' = L\dot{\theta} + r\Omega\cos(\Omega t - \theta)$$

We can finally compute the viscous friction force:

$$F_{\rm visc} = -k\left[L\dot{\theta} + r\Omega\cos(\Omega t - \theta)\right]. \tag{8}$$

Finally we have the inertial force due to the motion of O', where we use (7) to compute it

$$\vec{F}_{\rm in} \cdot \vec{e}_\theta = m\,(\vec{a}_{O'})_\theta = mr\Omega^2\sin(\Omega t - \theta) \tag{9}$$

Applying Newton's second law along the tangential direction, and plugging all the tangential forces we have found (4), (8) and (9), we get:

$$mL\ddot{\theta} = -mg\sin\theta - \kappa\left(L\dot{\theta} + r\Omega\cos(\Omega t - \theta)\right) + mr\Omega^2\sin(\Omega t - \theta)$$

Dividing by mL, we obtain:

$$\ddot{\theta} = -\frac{g}{L}\sin\theta - \frac{\kappa}{m}\left(\dot{\theta} + \frac{r\Omega}{L}\cos(\Omega t - \theta)\right) + \frac{r\Omega^2}{L}\sin(\Omega t - \theta)$$

Finally, the system can be written in the following form:

$$\frac{d}{dt}\begin{pmatrix}\theta \\ \dot{\theta}\end{pmatrix} = \begin{pmatrix} f_1(\theta, \dot{\theta}, t) \\ f_2(\theta, \dot{\theta}, t)\end{pmatrix}$$

with

$$f_1(\theta, \dot{\theta}, t) = \dot{\theta}$$

and

$$\boxed{f_2(\theta, \dot{\theta}, t) = -\frac{g}{L}\sin\theta - \frac{\kappa}{m}\left(\dot{\theta} + \frac{r\Omega}{L}\cos(\Omega t - \theta)\right) + \frac{r\Omega^2}{L}\sin(\Omega t - \theta)} \tag{10}$$

2.2 Mechanical energy and non-conservative power (b)

The mechanical energy in the frame $\mathcal{R}'$ is given by

$$E_M = T' + U'. \tag{11}$$

The kinetic energy is

$$T' = \frac{1}{2}m\vec{v}'^2 = \frac{1}{2}mL^2\dot{\theta}^2.$$

Taking $y = -L\cos\theta$, the potential energy is

$$U' = mgy = -mgL\cos\theta.$$

Thus, the mechanical energy is

$$\boxed{E_M = \frac{1}{2}mL^2\dot{\theta}^2 - mgL\cos\theta} \tag{12}$$

Using $P = \vec{F}\cdot\vec{v}$, the non-conservative forces are the friction force and the inertial force. The power of the friction force is

$$P_{\text{fric}} = -\kappa L\dot{\theta}\left(L\dot{\theta} + r\Omega\cos(\Omega t - \theta)\right).$$

The power of the inertial force is

$$P_{\text{in}} = mr\Omega^2 L\dot{\theta}\sin(\Omega t - \theta).$$

Therefore, the total power of non-conservative forces is

$$\boxed{P = -\kappa L^2\dot{\theta}^2 - \kappa r\Omega L\dot{\theta}\cos(\Omega t - \theta) + mr\Omega^2 L\dot{\theta}\sin(\Omega t - \theta)} \tag{13}$$

2.3 Solution for small-oscillation solution (c)

We consider the case without damping ($\kappa = 0$) and without excitation ($r = 0$). In this situation, the equation of motion derived in the first section reduces to

$$\ddot{\theta} = -\frac{g}{L}\sin\theta.$$

For small oscillations ($\theta \ll 1$), we use the linear approximation $\sin\theta \approx \theta$, which leads to

$$\ddot{\theta} + \frac{g}{L}\theta = 0.$$

This is the equation of a simple harmonic oscillator with natural frequency

$$\omega_0 = \sqrt{\frac{g}{L}} \tag{14}$$

The general solution is therefore given by

$$\theta(t) = A\cos(\omega_0 t) + B\sin(\omega_0 t),$$

where A and B are constants that can be determined by the initial conditions. Assuming $\theta(0) = \theta_0$ and $\dot{\theta}(0) = \dot{\theta}_0$, we determine the constants.

$$\theta(0) = A\cos(\omega_0 \cdot 0) = \theta_0 \Rightarrow A = \theta_0$$

$$\dot{\theta}(0) = B\omega_0 \cos(\omega_0 \cdot 0) = \dot{\theta}_0 \Rightarrow B = \frac{\dot{\theta}_0}{\omega_0}$$

Finally the solution will be

$$\boxed{\theta(t) = \theta_0 \cos(\omega_0 t) + \frac{\dot{\theta}_0}{\omega_0} \sin(\omega_0 t)} \tag{15}$$

And in the particular case where $\dot{\theta}_0 = 0$, this simplifies to

$$\theta(t) = \theta_0 \cos\left(\sqrt{\frac{g}{L}}\, t\right).$$

3 Numerical method and implementation

3.1 Equation of motion and state variables

The motion of the pendulum is described in terms of the angle $\theta(t)$ and angular velocity $\dot{\theta}(t)$. The problem is written as a first-order system,

$$\frac{d}{dt}\begin{pmatrix}\theta \\ \dot{\theta}\end{pmatrix} = \begin{pmatrix} f_1(\theta, \dot{\theta}, t) \\ f_2(\theta, \dot{\theta}, t)\end{pmatrix}, \tag{16}$$

with

$$f_1(\theta, \dot{\theta}, t) = \dot{\theta},$$

while $f_2(\theta, \dot{\theta}, t)$ is the angular acceleration derived analytically in Section 2. In the code, the variables θ and $\dot{\theta}$ are stored as the dynamical unknowns and are updated at each time step. The physical parameters of the problem are the mass m, pendulum length L, gravitational acceleration g, forcing radius r, forcing angular frequency Ω, and damping coefficient κ.
This formulation is convenient since it gives direct access both to the trajectory $\theta(t)$ and to the phase-space representation $(\theta, \dot{\theta})$, which is essential for the study of resonance, sensitivity to initial conditions and Poincaré sections.

3.2 Velocity-Verlet scheme for a non-autonomous dissipative system

The exercise requires the implementation of a second-order Verlet scheme, adapted to the case where the acceleration depends explicitly on position, velocity and time. Contrary to the standard conservative case, the present acceleration is not a function of θ alone, because both viscous damping and external forcing contribute through $\dot{\theta}$ and t.
Denoting the angular acceleration by

$$a(\theta, \dot{\theta}, t) = f_2(\theta, \dot{\theta}, t),$$

the numerical scheme used in the code is written in predictor-corrector form. Starting from $(\theta_n, \dot{\theta}_n)$ at time t_n, we first predict the new angular position through

$$\theta_{n+1} = \theta_n + \Delta t\, \dot{\theta}_n + \frac{1}{2}\Delta t^2\, a(\theta_n, \dot{\theta}_n, t_n). \tag{17}$$

A provisional half-step velocity is then introduced,

$$\dot{\theta}_{n+\frac{1}{2}} = \dot{\theta}_n + \frac{1}{2}\Delta t\, a(\theta_n, \dot{\theta}_n, t_n), \tag{18}$$

and the angular velocity is finally corrected using the accelerations evaluated at the old and new times,

$$\dot{\theta}_{n+1} = \dot{\theta}_n + \frac{1}{2}\Delta t \left[a(\theta_n, \dot{\theta}_{n+\frac{1}{2}}, t_n) + a(\theta_{n+1}, \dot{\theta}_{n+\frac{1}{2}}, t_{n+1}) \right] . \tag{19}$$

This approach preserves the spirit of the velocity-Verlet method while allowing for the explicit dependence of the force on velocity and time. It provides a second-order accurate integration scheme, which is particularly suitable for long-time simulations where first-order methods would accumulate significantly larger numerical errors.

3.3 Mechanical energy and non-conservative power

Besides the trajectory itself, the code also computes two diagnostic quantities requested in the statement: the mechanical energy of the system and the power associated with non-conservative forces.

The mechanical energy is evaluated from the kinetic and gravitational potential contributions of the pendulum. In the absence of forcing and damping, this quantity should remain constant up to numerical errors, which makes it a useful tool for validating the implementation in the simple reference case $r = 0$ and $\kappa = 0$.

The power of the non-conservative forces includes both viscous dissipation and the contribution associated with the inertial forcing induced by the circular motion of the support. Monitoring this quantity is useful to characterise energy exchange with the environment in the forced and damped regime, where the total mechanical energy is no longer conserved.

3.4 Time discretisation and simulation parameters

All simulations are performed on a uniform time grid. Once the final time t_f and the number of time steps `nsteps` are fixed, the time step is defined by

$$\Delta t = \frac{t_f}{\texttt{nsteps}}, \tag{20}$$

except in the cases where the simulation is prescribed over a fixed number of forcing periods N, in which case the final time is set by

$$t_f = N \frac{2\pi}{\Omega},$$

and the time step becomes the forcing period divided by the number of steps per period.

This implementation makes it possible to adapt the temporal resolution to the physical regime being studied. Short simulations with moderate time resolution are sufficient for the convergence study of the small-oscillation problem, while much longer runs are required for the study of resonance, chaotic trajectories and Poincaré sections.

3.5 Outputs and post-processing

At each sampled time step, the program writes to the output file the current values of time, angular position, angular velocity, mechanical energy and non-conservative power. The output format is therefore

$$(t, \theta, \dot{\theta}, E_{\mathrm{mec}}, P_{\mathrm{nc}}).$$

These quantities are then post-processed in Python in order to build the figures presented in Section 4.

Depending on the subsection considered, the post-processing is used to determine the oscillation period from the numerical trajectory, compare the numerical solution with the analytical one, compute the maximum amplitude reached during forced oscillations, measure the distance between nearby trajectories in chaotic and non-chaotic cases, and construct Poincaré sections by interpolation at times $t = nT$, where $T = 2\pi/\Omega$ is the forcing period.

4 Simulations and analyses

4.1 (a) Small-oscillation regime: convergence and energy conservation

In this first set of simulations, we consider the simplest limit of the problem, namely the unforced and undamped pendulum,

$$r = 0, \qquad \kappa = 0,$$

in the small-oscillation regime. In this case, the equation of motion reduces to the harmonic approximation discussed in Section 2, for which the analytical solution is known explicitly. This configuration is therefore used as a reference case to validate the numerical implementation.
The simulations were performed with

$$m = 0.1, \qquad L = 0.2, \qquad g = 9.81,$$

and initial conditions

$$\theta(0) = 10^{-3}, \qquad \dot{\theta}(0) = 0.$$

The final time was fixed to $t_f = 1.8$, corresponding to about two oscillation periods, and the number of time steps was varied from 32 to 4096. In this way, the time step

$$\Delta t = \frac{t_f}{\texttt{nsteps}}$$

was progressively reduced in order to study the convergence of the numerical solution.

Comparison with the analytical solution. Figure 1 compares the numerical trajectories obtained for three representative discretisations with the analytical solution. As expected, the coarsest simulation (`nsteps`=32) already reproduces the oscillatory behaviour correctly, but small deviations with respect to the exact solution are visible. When the time step is reduced, the agreement improves significantly, and for the finest runs the numerical and analytical curves are almost indistinguishable on the scale of the figure. This provides a first visual confirmation that the implemented scheme converges toward the exact solution.
To quantify this convergence, we computed the maximum trajectory error

$$\varepsilon_\theta = \max_t |\theta_{\text{num}}(t) - \theta_{\text{ana}}(t)|$$

for each value of Δt. The result is shown in Figure 2. The error decreases monotonically as the time step is refined, showing that the numerical trajectory converges toward the analytical one. Moreover, the decrease is clearly faster than a first-order reference scaling $\mathcal{O}(\Delta t)$, which is consistent with the good accuracy expected from the Verlet-type scheme implemented here.

Numerical period. A second validation consists in extracting the oscillation period from the numerical trajectories and comparing it with the analytical period of the linearised pendulum,

$$T_0 = 2\pi\sqrt{\frac{L}{g}}.$$

The numerical period was estimated from the zero crossings of $\theta(t)$, and the relative error

$$\varepsilon_T = \frac{|T_{\text{num}} - T_0|}{T_0}$$

was then plotted as a function of Δt. As shown in Figure 3, the relative error on the period also decreases regularly when the time step is reduced. This confirms that the numerical scheme not only reproduces the full analytical trajectory, but also correctly captures the fundamental physical observable associated with the motion, namely the oscillation period.

Mechanical energy conservation. Since $r = 0$ and $\kappa = 0$, the system is conservative and the mechanical energy should remain constant. For the initial conditions used here, the initial angular velocity is zero, so the mechanical energy is purely potential:

$$E_{\text{mec}}(0) = mgL(1 - \cos\theta_0).$$

With $\theta_0 = 10^{-3}$, this gives

$$E_{\text{mec}}(0) \approx 9.81 \times 10^{-8} \text{ J},$$

which is precisely the value around which the numerical energy evolves.
Figure 4 shows the mechanical energy for the finest simulation. The energy remains extremely close to a constant value over the whole integration interval, with only tiny oscillations around

$$E_{\text{mec}} \approx 9.81 \times 10^{-8} \text{ J}.$$

Using the numerical output, we obtain

$$E_{\min} = 9.80997595018696 \times 10^{-8} \text{ J}, \qquad E_{\max} = 9.80999918321121 \times 10^{-8} \text{ J},$$

so that the total variation is of order

$$E_{\max} - E_{\min} \sim 10^{-13} \text{ J},$$

corresponding to a relative fluctuation of order 10^{-6}. This very small drift confirms that the conservative dynamics is accurately reproduced and that the numerical implementation is reliable in this reference regime.
Overall, these results validate the code in the simplest case where analytical predictions are available: the trajectory converges toward the exact solution, the numerical period converges toward the theoretical one, and the mechanical energy is conserved up to very small discretisation errors.

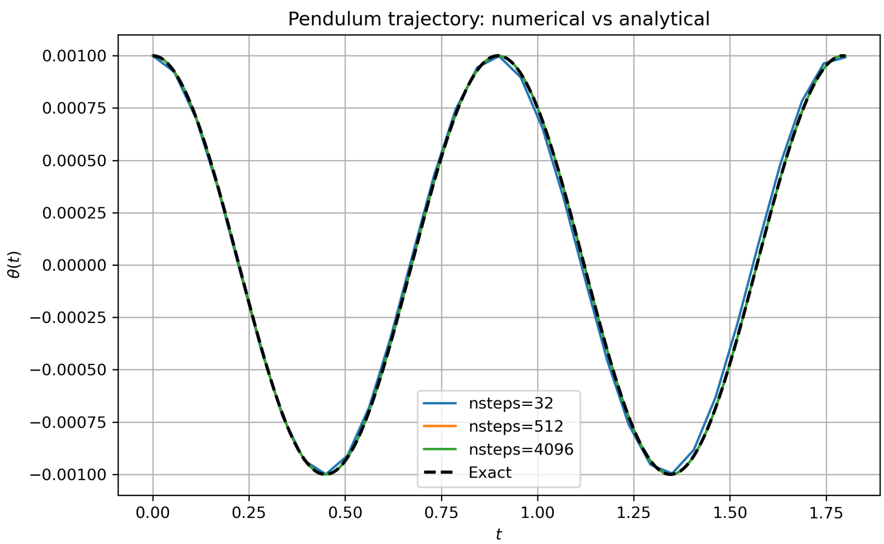

Figure 1: Comparison between the analytical small-oscillation solution and numerical trajectories obtained with different time steps.

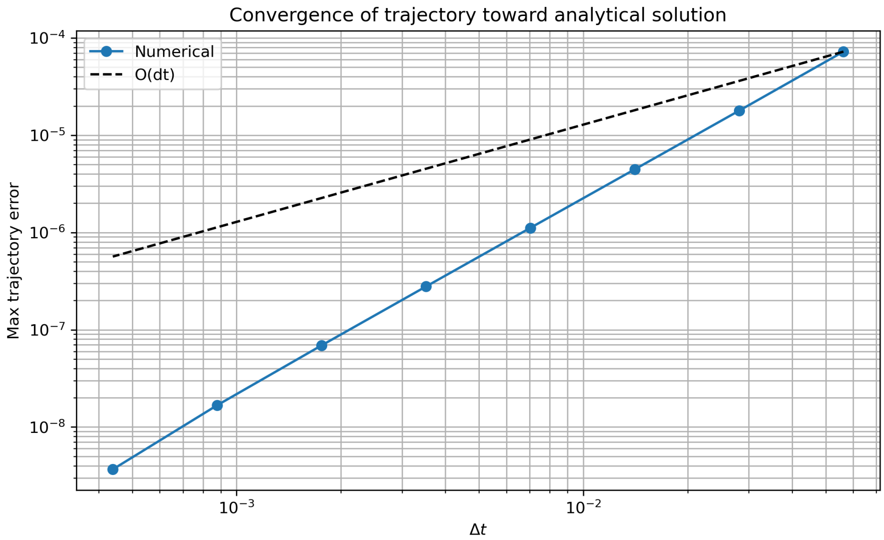

Figure 2: Maximum trajectory error as a function of the time step Δt.

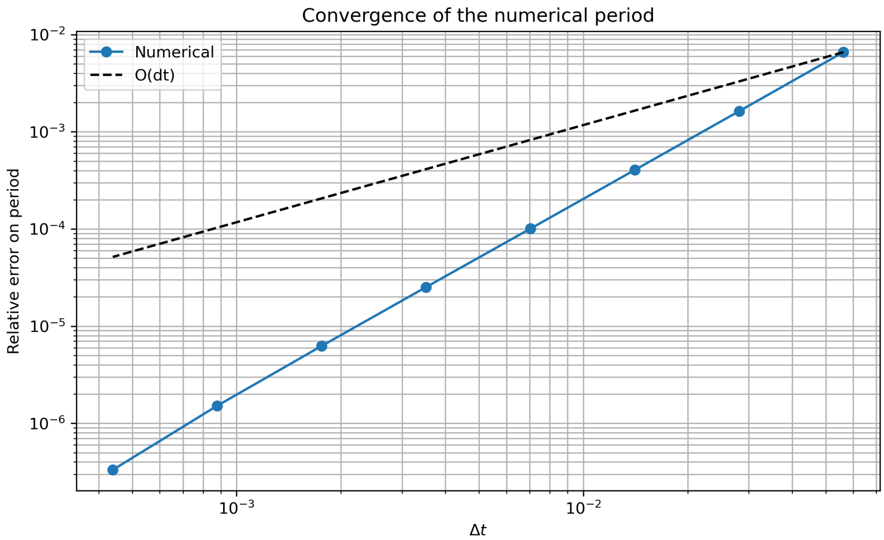

Figure 3: Relative error on the numerical oscillation period as a function of Δt.

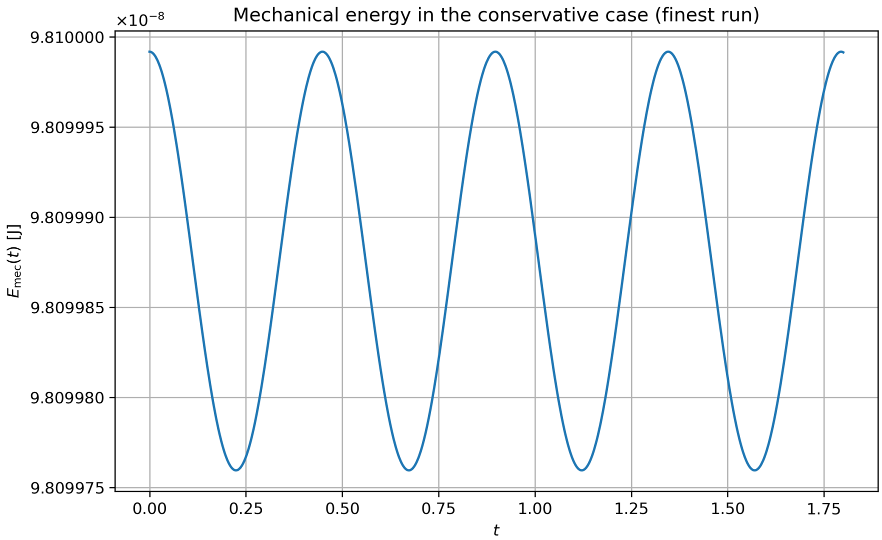

Figure 4: Mechanical energy in the conservative case for the finest simulation.

4.2 (b) Dependence of the period on the oscillation amplitude

In this second part, we still consider the unforced and undamped pendulum,

$$r = 0, \qquad \kappa = 0,$$

but we now move beyond the small-oscillation regime studied in part (a). In the linear approximation, obtained from $\sin\theta \approx \theta$, the period is independent of the initial amplitude and is simply given by

$$T_0 = 2\pi\sqrt{\frac{L}{g}}.$$

However, for the full nonlinear pendulum, the period depends on the oscillation amplitude θ_0, and increases as the initial angle becomes larger.
To investigate this effect numerically, we fixed the physical parameters

$$m = 0.1, \qquad L = 0.2, \qquad g = 9.81,$$

and used the initial conditions

$$\theta(0) = \theta_0, \qquad \dot{\theta}(0) = 0,$$

where θ_0 was varied from very small values up to values close to π. The time step was kept sufficiently small (`nsteps`=4096) so that the numerical error due to time discretisation remained negligible compared to the physical amplitude effect under study.

Analytical nonlinear period. For the nonlinear pendulum, the exact analytical expression of the period is

$$T(\theta_0) = 4\sqrt{\frac{L}{g}}\, K\left(\sin^2\frac{\theta_0}{2}\right),$$

where K denotes the complete elliptic integral of the first kind. This formula reduces to the small-amplitude result T_0 when $\theta_0 \to 0$, but departs from it as the amplitude increases. In particular, when $\theta_0 \to \pi$, the period diverges, since the pendulum starts arbitrarily close to the unstable upper equilibrium and therefore takes an increasingly long time to leave that region.

Comparison between numerical and analytical results. Figure 5 shows the numerical period as a function of the initial amplitude together with the analytical nonlinear prediction. The agreement is very good over the whole interval explored. For small amplitudes, the period remains close to the constant small-oscillation value, which is consistent with the linear theory discussed in part (a). As θ_0 increases, the numerical period also increases, showing the expected nonlinear behaviour. Close to $\theta_0 = \pi$, the growth becomes much steeper, in agreement with the divergence predicted by the analytical formula.
To quantify the quality of the numerical result, we also computed the relative error between the simulated and analytical periods,

$$\varepsilon_T(\theta_0) = \frac{|T_{\text{num}} - T_{\text{ana}}|}{T_{\text{ana}}}.$$

The corresponding curve is shown in Figure 6. The relative error remains small over the whole range of amplitudes, but the figure shows noticeable point-to-point fluctuations. These irregular variations are attributed to the numerical extraction of the period from discretely sampled trajectories: since the period is estimated from the detected positions of successive maxima of $\theta(t)$, small shifts in the localisation of these maxima on the time grid can produce visible changes in the computed error.
Overall, this part confirms that the numerical model captures the transition from the linear regime, where the period is essentially constant, to the nonlinear regime, where the period depends strongly on the oscillation amplitude.

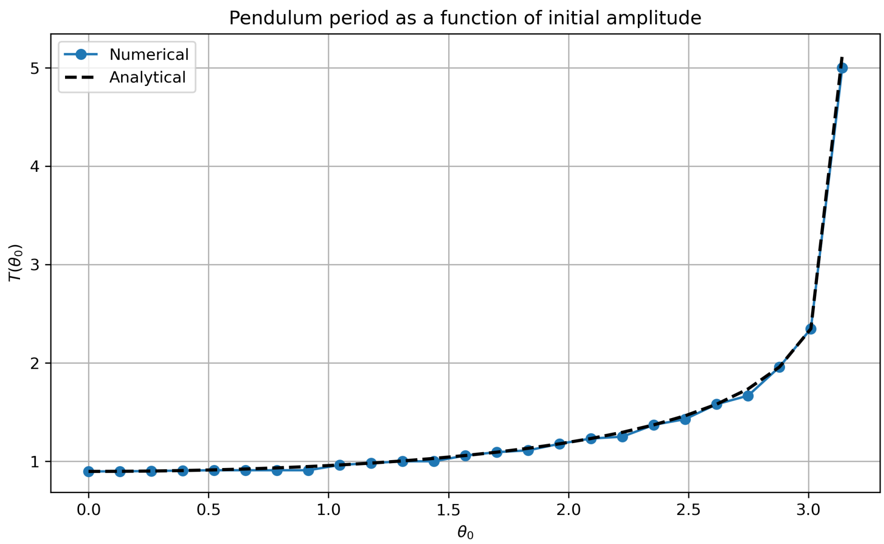

Figure 5: Pendulum period as a function of the initial amplitude θ_0. The numerical results are compared with the analytical nonlinear prediction.

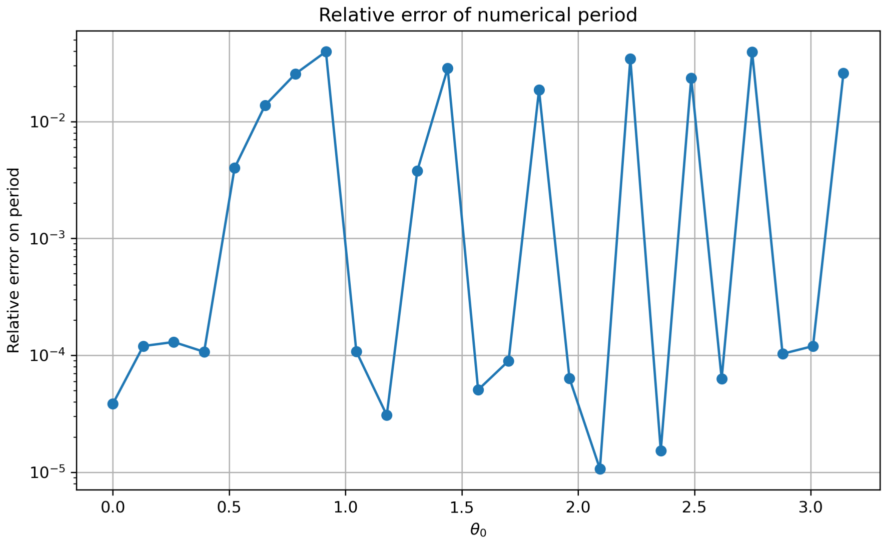

Figure 6: Relative error between the numerical and analytical periods as a function of θ_0.

4.3 (c) Resonance, sensitivity to initial conditions, and transition to chaos

4.3.1 (i) Resonance

In order to find the maximum amplitude we perform some simulations of 50s with $r = 0.01$, $\kappa = 0.02$, $\theta_0 = 0$ and $\dot{\theta}_0 = 0$, and changing the frequency from 5 to 9 at intervals of 0.1. For each value of Ω, we measure the maximum amplitude reached by the system, defined as $\max |\theta(t)|$. The resulting curve of $\max |\theta|$ as a function of Ω is shown at Figure 7 and it exhibits a clear resonance peak.

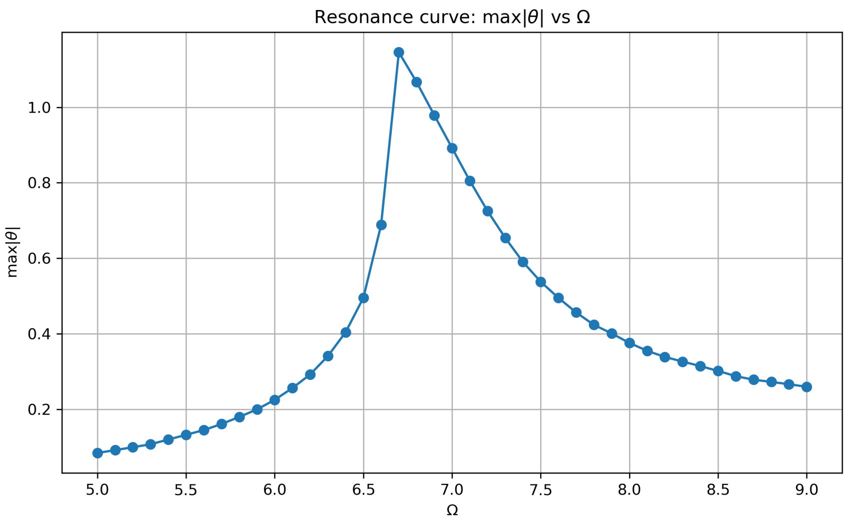

Figure 7: $\max |\theta|$ as a function of Ω

The maximum amplitude we obtained is

$$\max |\theta| = 1.146$$

for a frequency of $\Omega = 6.7\,\text{rad/s}$.

As it should be, the value is of the resonance peak is finite and near the natural frequency of the pendulum in the small-angle approximation, given by (14)

$$\omega_0 = \sqrt{\frac{g}{L}} \approx 7.0\,\mathrm{rad/s}.$$

Moreover, the resonance peak is slightly shifted towards lower frequencies due to the damping term ($\kappa \neq 0$). For higher frequencies, we see that the amplitude decreases again since the system is no longer able to follow the external driving, and it gets far from the resonance value. Overall, the observed behavior is characteristic of a damped system exhibiting resonance.

4.3.2 (ii) Movement types: chaotic and periodic

For this section we have done numerous simulations considering a time interval of 100 s, with fixed parameters $\Omega = \sqrt{g/L}$, $\kappa = 0.01$, and initial condition $\theta_0 = 0$. The forcing amplitude r was varied between 0 and 0.1. For each of these values of r we have plotted a phase space diagram, Figure 8, where we can really analyze the chaos and the different types of movements observed.

We have found that for small values of r, from 0 to 0.05, the system exhibits a periodic movement. This is observed in the phase space diagram, where trajectories form closed curves and clean patterns, indicating that the system returns to the same state after each period.

To be more precise, as r increases, the motion becomes quasi-periodic. In this regime, trajectories no longer lie on a single circular closed curve but instead form a torus-like structure in the phase space diagram. A good example of this would be the phase diagram for $r = 0.05$

For larger values of r, from 0.06 to 0.1, the system enters a chaotic movement. The trajectories in phase space become irregular and fill a region rather than following a well-defined curve. There is not a clear pattern, especially for larger times, and no closed curve either. There is a clear difference in shape between the first row diagrams and the ones in the second row.

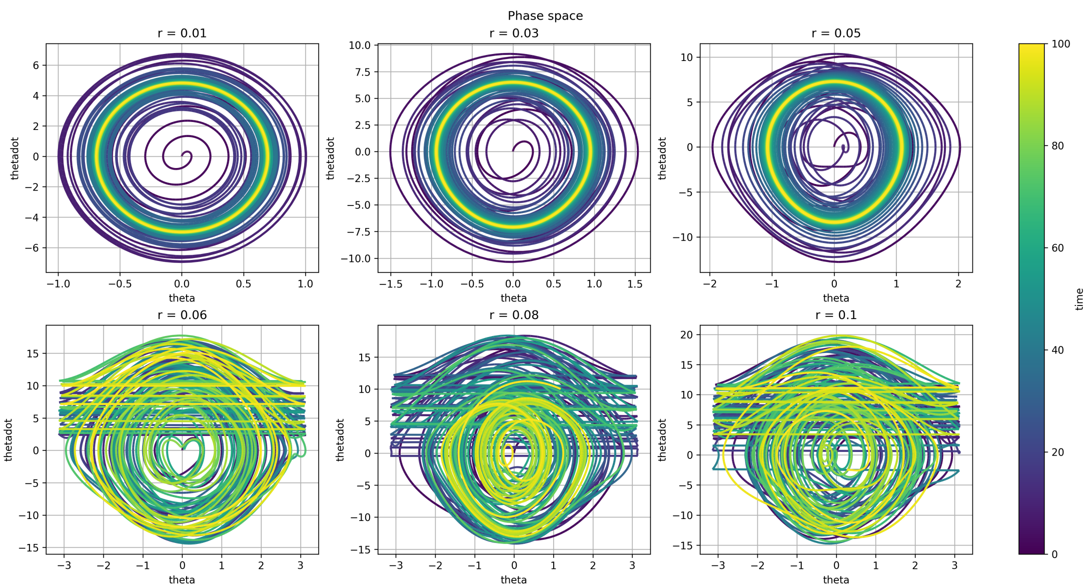

Figure 8: Phase space trajectories for different values of r

We see that increasing the radius r leads to a transition from periodic to chaotic movement. Thus, increasing the radius leads to an increase of the chaos of the system. To better confirm our observations from the phase space diagram, we have also plotted the angular position as a function of time.

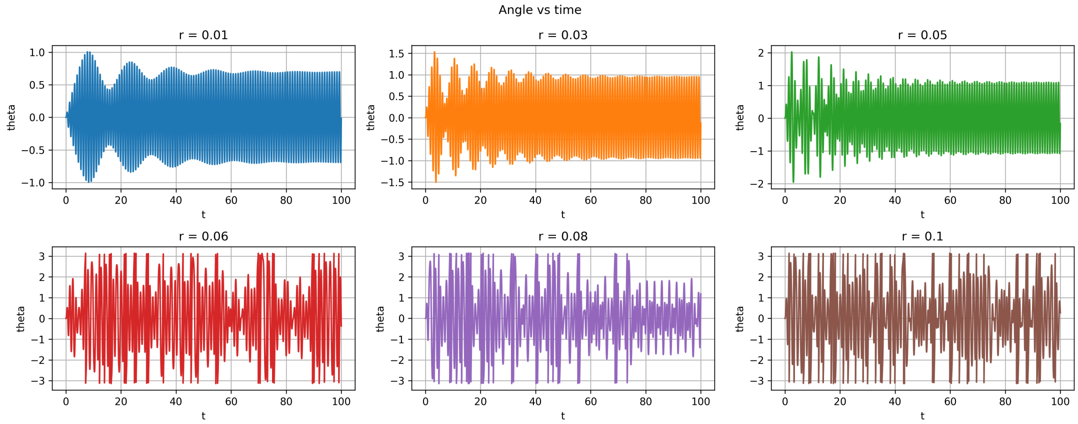

Figure 9: theta as a function of time for different values of r

In Figure 9, we observe the same qualitative behavior as discussed previously. For small values of r, corresponding to the periodic or quasi-periodic regime, the angle $\theta(t)$ evolves towards a regular oscillatory motion with a well-defined structure. Although it does not converge to a constant value, the dynamics remain bounded and exhibit a clear pattern over time.

On the other hand, for larger values of r (second row), the system enters a chaotic regime. In this case, $\theta(t)$ no longer exhibits regular oscillations and instead displays irregular, aperiodic behavior. The signal remains bounded but shows no apparent repeating pattern.

The observed transition between regular and chaotic dynamics can be understood from the analytical structure of the equation of motion (10). The system is governed by a nonlinear second-order differential equation with damping and periodic forcing. The presence of the $\sin(\theta)$ term introduces nonlinearity, while the external forcing with frequency Ω continuously injects energy into the system, and the damping term $\kappa\dot{\theta}$ dissipates it. For small values of the forcing amplitude r, the system behaves close to a linear oscillator, and the solution remains regular, leading to periodic or quasi-periodic motion. This explains the structured and predictable oscillations observed in the first row of Figure 9 or Figure 8.

However, as r increases, the nonlinear effects become dominant and the balance between energy injection and dissipation leads to complex dynamics. In this regime, the system undergoes a loss of stability of regular motion, and small perturbations are amplified over time. This results in aperiodic, irregular behavior, characteristic of chaos, as observed in the second row of the figure. Although no exact analytical solution exists in this regime, the qualitative behavior is well understood: the system evolves towards a chaotic attractor, which explains the bounded yet unpredictable nature of $\theta(t)$. We will discuss further this notion later on the report.

4.3.3 (iii) Sensitivity to initial conditions in chaotic and non-chaotic regimes

To further study the chaotic movement, two simulations were performed with nearly identical initial conditions, differing only by a small perturbation in the initial conditions, $\Delta\theta_0 = 10^{-10}$. To analyse the implications of this perturbation, we computed the time evolution of the difference between the two trajectories, $|\Delta\theta(t)|$ and plotted it in Figure 10. There are 2 trajectories in the plot, one for each of the movement types: chaotic or non chaotic.

For a value of r corresponding to a chaotic regime, $r = 0.08$, the separation between the two trajectories starts growing exponentially over time until it stabilizes. When plotted on a semi-logarithmic scale, $|\Delta\theta(t)|$ has an approximately linear growth at the start, indicating an exponential divergence of nearby trajectories.

In contrast, for a value of r corresponding to a non chaotic (periodic or quasi-periodic) regime, $r = 0.01$, the separation between the two trajectories remains bounded. The $|\Delta\theta(t)|$ starts with a very small value and it has a very small decrease of it over time. Indeed, showing, in this regime, how little impact a slight change in the initial conditions has on their trajectories over time.

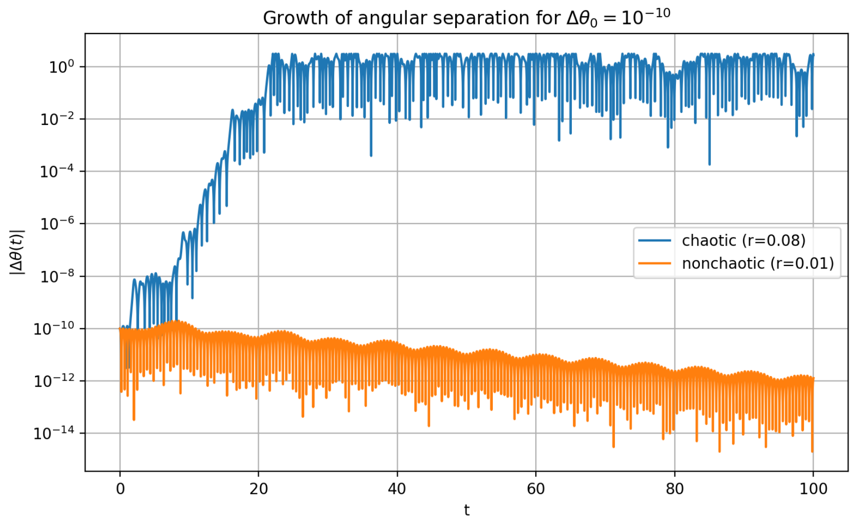

Figure 10: Lin-log diagram of the evolution of the $|\Delta\theta(t)|$ over time for a chaotic and a non chaotic behaviours

Comparing the chaotic regime to the non chaotic regime we see that it has a higher $|\Delta\theta(t)|$ all the time. In addition, it also has a bigger growth of the angular separation since the start of the simulation. Therefore, these results clearly demonstrate that chaotic dynamics are characterized by exponential sensitivity to initial conditions. On the other hand, the system in the non chaotic regime is stable with respect to small perturbations in the initial conditions, and also, it does not exhibit a exponential trend on the semi-logarithmic plot.

This behaviour can be traced back to the structure of phase space that we saw before. In regular regimes, motion is confined to invariant curves (closed or quasi-periodic trajectories), which prevent trajectories from diverging significantly. In chaotic regimes, these structures break down, allowing trajectories to explore a larger region of phase space and leading to exponential separation.

4.3.4 (iv) Optional: Numerical Convergence and chaos

To study the numerical convergence for the two different types of movement, we did several simulations using different but sufficiently small time steps Δt.

As for the non-chaotic regime, the trajectories obtained with different time steps remain exactly the same over the entire simulation time. This indicates that the numerical method converges properly and that the solution is stable with respect to small changes in Δt. We can see this in the following Figure 11.

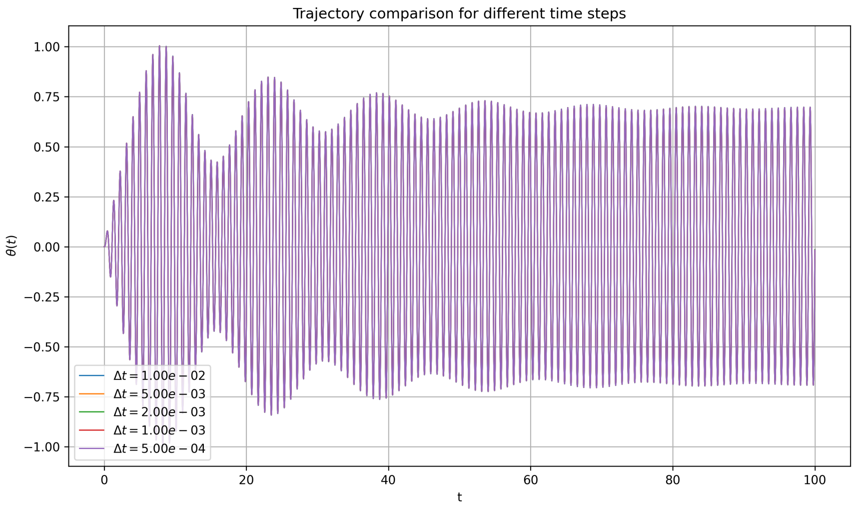

Figure 11: Trajectory comparison for different time steps in the non chaotic regime

On the other hand, for the chaotic movement, the trajectories initially coincide but progressively diverge over time, even when very small time steps are used. We see this in Figure 12 when very early on in the simulation, barely $t = 9s$, the trajectories from different time steps start diverging from each other and end up adopting totally different trajectories.

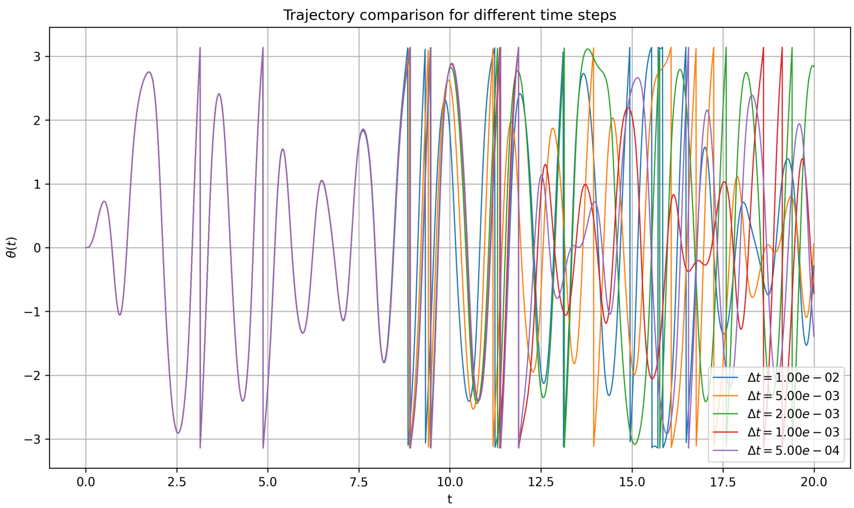

Figure 12: Trajectory comparison for different time steps in the chaotic regime

This divergence is not due to a lack of numerical convergence, but rather reflects the intrinsic sensitivity of chaotic systems to small perturbations. In this context, the difference in Δt acts as a small perturbation that grows exponentially as we saw before. This is why they start diverging so early on in the simulation. Therefore, the numerical method can be considered convergent in a short time sense, but long-term agreement between trajectories cannot be expected in the chaotic regime due to the nature of this type of movement.

4.4 (d) Poincaré sections and strange attractors

For this section, we did long-time simulations of the system over several thousand excitation periods in order to analyze its asymptotic behavior. The Poincaré section was constructed by sampling the phase-space trajectory $(\theta, \dot{\theta})$ at times $t = nT$, where $T = 2\pi/\Omega$ is the forcing period. To improve accuracy, θ and $\dot{\theta}$ at these times were obtained by interpolation.

We considered three representative values of r:

- $r = 0.01$, corresponding to a weak forcing regime,
- $r = 0.08$, corresponding to a chaotic regime.
- $r = 0.1$, corresponding to a stronger forcing regime.

The resulting Poincaré sections are shown in Figure 13. For the small forcing amplitude $r = 0.01$, the Poincaré section is confined to a very thin and structured set in phase space. This indicates a regular regime, where the motion remains close to a stable attractor. The narrow distribution suggests that the dynamics is either periodic or quasi-periodic, with very small amplitude oscillations around equilibrium.

In contrast, for the larger forcing amplitude $r = 0.08$ or $r = 0.1$, the Poincaré section spreads over a broad region of phase space, forming a scattered cloud of points. This behavior is characteristic of a chaotic regime, where the trajectory does not settle onto a simple closed orbit or smooth curve. Instead, the system explores a large portion of phase space in an apparently irregular manner. In particular, the extended and irregular structure observed is consistent with the presence of a strange attractor.

Periodic motion leads to a finite number of isolated points, quasi-periodic motion to smooth closed curves, and chaotic motion to a scattered fractal-like set corresponding to a strange attractor, like for $r = 0.08$ or $r = 0.1$. In the chaotic regime, the Poincaré section does not reduce to isolated points or smooth curves, but instead forms a complex set with an irregular structure. This set is known as a strange attractor: it reflects deterministic dynamics while exhibiting sensitivity to initial conditions and a fractal-like geometry. The qualitative difference between these two cases illustrates the transition from regular to complex dynamics as r increases.

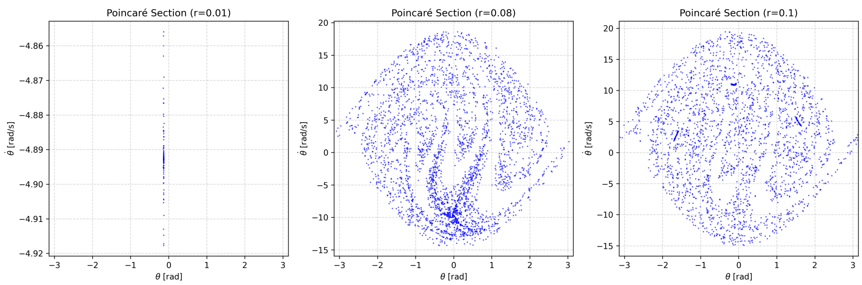

Figure 13: Poincare sections for different r

4.5 (e) Other attractors

As suggested in the optional part of the statement, we further explored the long-time dynamics of the system by varying both the forcing amplitude r and the driving frequency Ω, in order to obtain different attractors in the Poincaré section. The goal of this part is twofold: first, to show that different choices of parameters may lead to qualitatively different asymptotic sets in phase space; second, to verify that for a given chaotic case, changing the initial condition does not significantly modify the final Poincaré section.
We used as a reference chaotic case the same type of parameters as in the previous subsection, corresponding to a regime where the Poincaré section fills a broad region of phase space. The resulting section is shown in Figure 15. Its scattered two-dimensional structure is characteristic of a chaotic attractor: the trajectory does not converge to a simple closed curve, but instead visits a large portion of the accessible phase space in an irregular yet bounded way.

For comparison, Figure 14 shows a reference regular case. In this situation, the Poincaré points remain concentrated on a very thin closed structure. This behaviour is consistent with regular dynamics, in contrast with the extended cloud observed in the chaotic regime. The comparison between Figures 14 and 15 illustrates clearly how the nature of the attractor changes when the system parameters are modified.

We then varied the driving frequency while keeping a chaotic-type regime, and obtained the Poincaré section shown in Figure 16. This section is still spread over a two-dimensional region of phase space, but its shape is visibly different from that of the reference chaotic attractor. In particular, the distribution of points is more concentrated in some regions, with a visible accumulation near $\theta \approx 0$ and $\dot{\theta} \approx -10$, and is less homogeneous than in Figure 15. This confirms that changing Ω can modify the geometry of the attractor, even when the dynamics remains irregular.

Finally, in order to test the robustness of the attractor with respect to the initial condition, we considered the same physical parameters as in the reference chaotic case, but used two different initial conditions. The corresponding Poincaré sections are superimposed in Figure 17. We observe that both sets of points occupy essentially the same region of phase space and reproduce the same global structure. Small local differences remain due to the finite simulation time and the fact that the attractor is sampled by discrete iterates, but the overall set is practically unchanged. This is precisely the behaviour expected for an attractor: trajectories starting from different initial conditions are drawn toward the same asymptotic set.

These results show that the system admits several qualitatively different attractors depending on the forcing parameters. Regular regimes lead to thin structured sets, while chaotic regimes lead to extended strange attractors. Moreover, for a fixed chaotic case, the Poincaré section remains essentially the same when the initial condition is varied, confirming that the observed structure is a genuine attractor of the dynamics rather than a transient feature of a particular trajectory.

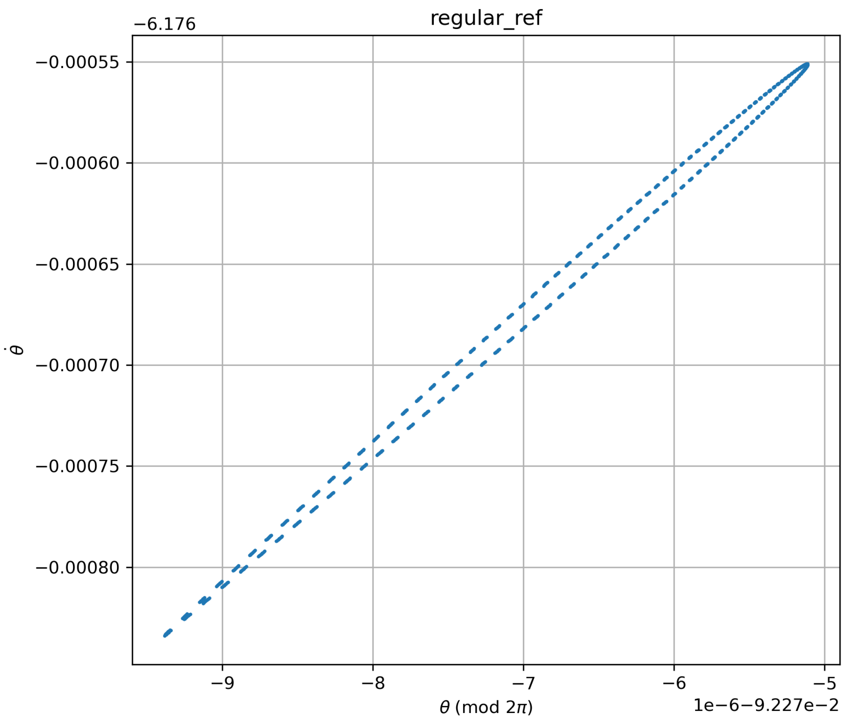

Figure 14: Reference regular attractor in the Poincaré section. The points remain confined to a thin closed structure, characteristic of regular dynamics.

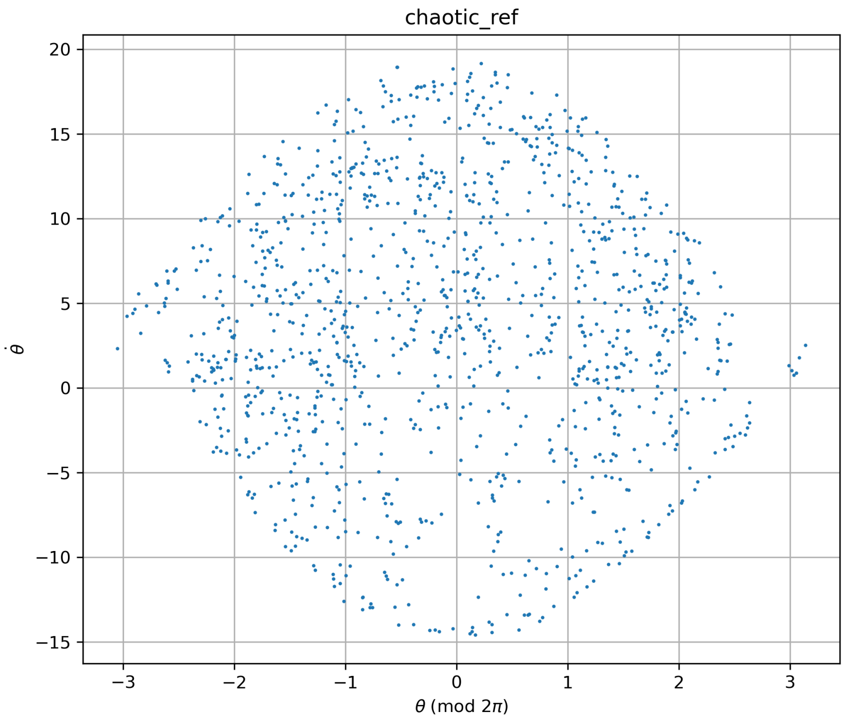

Figure 15: Reference chaotic attractor in the Poincaré section. The points fill a broad bounded region of phase space, which is characteristic of chaotic dynamics.

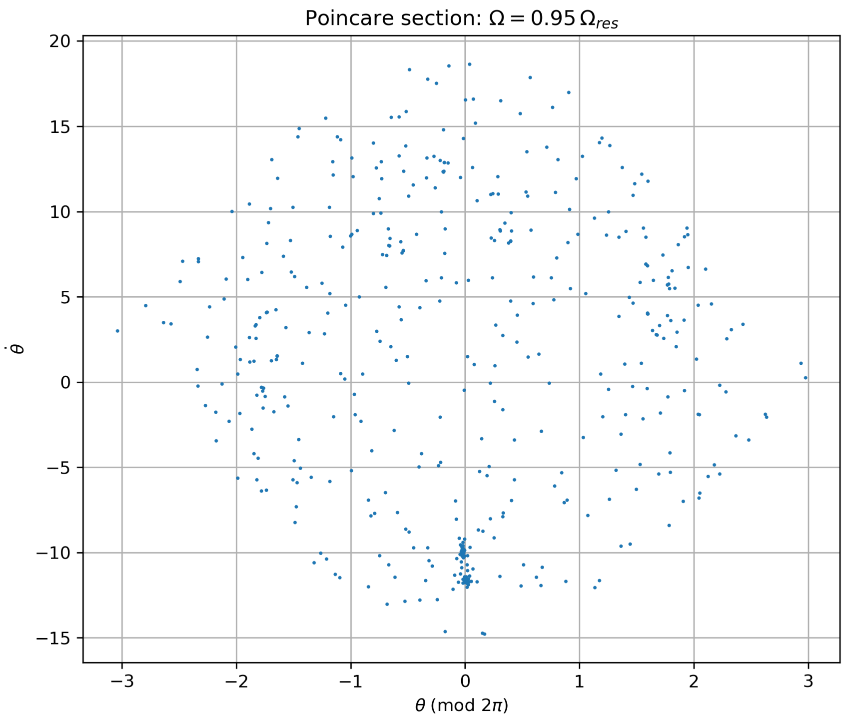

Figure 16: Poincaré section obtained after modifying the driving frequency to $\Omega = 0.95\,\Omega_{\mathrm{res}}$. The attractor remains irregular, but its geometry differs from that of the reference chaotic case.

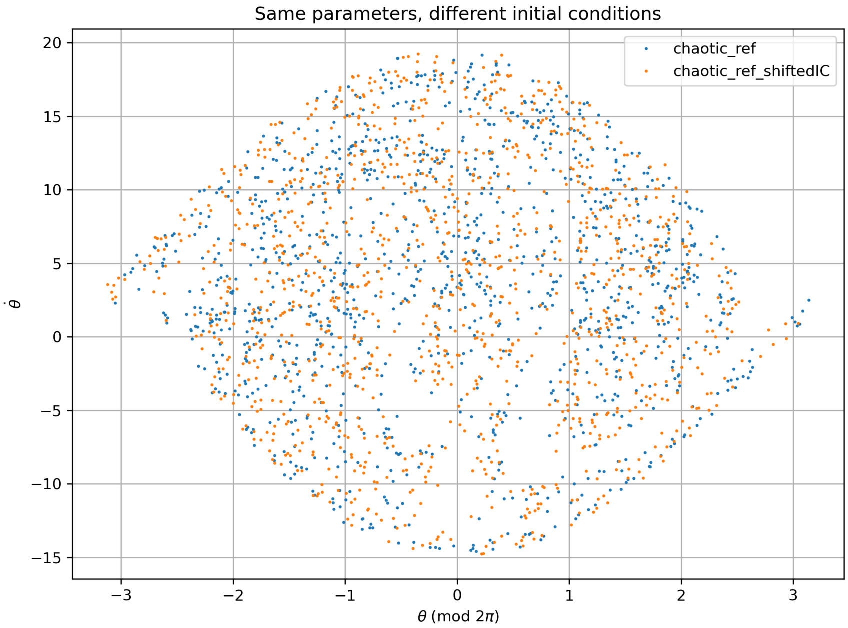

Figure 17: Superposition of two Poincaré sections obtained with the same physical parameters but different initial conditions. The two trajectories approach essentially the same asymptotic attractor.

5 Conclusion

In this work, we studied the dynamics of a pendulum with circular excitation, damping and nonlinear effects, combining analytical results with numerical simulations. In the small-oscillation limit, the numerical scheme was successfully validated by comparison with the analytical solution: the trajectory and the period converged correctly, and the mechanical energy remained nearly constant in the conservative case.

We then showed that beyond the linear regime, the oscillation period depends on the initial amplitude, in very good agreement with the nonlinear analytical prediction. In the forced and damped case, the simulations revealed a clear resonance peak near the natural frequency of the pendulum, slightly shifted due to damping.

By increasing the forcing amplitude, we observed a transition from regular to chaotic dynamics. This transition was confirmed through phase-space trajectories, time series, and sensitivity to initial conditions. In particular, nearby trajectories diverged rapidly in the chaotic regime, while they remained close in the regular regime.

Finally, the Poincaré sections provided a clear visual characterisation of the long-time dynamics. Regular regimes produced thin structured sets, whereas chaotic regimes led to extended strange attractors. We also showed that different parameter choices may generate different attractors, and that for a fixed chaotic case, changing the initial condition leaves the Poincaré section essentially unchanged.

Overall, the exercise illustrates how a simple nonlinear forced pendulum can display a wide variety of behaviours, ranging from harmonic motion to resonance and chaos, and shows that the implemented numerical method is able to capture all these regimes reliably.

6 References

1. Exercise statement: *Physique Numérique – Exercice 2, Pendule avec excitation circulaire, amortissement, résonance, chaos.*

2. L. Villard, *Physique Numérique*, sections on Verlet schemes and oscillating systems.

3. Use of LLM tools (ChatGPT) for language correction and report rephrasing.